Exp No:11 Experiment to understand KNN algorithm for a given data set

CODE

```
import numpy as np

import pandas as pd

df=pd.read_csv('Iris.csv')

df.info()

df.variety.value_counts()

df.head()

features=df.iloc[:,:-1].values

label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,ra
ndo

model_KNN=KNeighborsClassifier(n_neighbors=5)

print(model_KNN.score(xtrain,ytrain))

print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix

confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report

print(classification_report(label,model_KNN.predict(features)))
```

OUTPUT

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

RESULT Hence KNN algorithm for the given data set has been analyzed.